

Chemical Investigation and Antinociceptive Activity Evaluation of *Marrubium Vulgare* L. Aqueous Extract

Abdelfatah Aitbaba,^[a] Hamid Kabdy,^[a] Abdelmounaim Baslam,^[a] Hajar Azraida,^[a] Rachida Aboufatima,^[b] Loubna El Yazouli,^[a] Zahra Sokar,^[a] Stefania Garzoli,^{*,[c]} and Abderrahman Chait^[a]

Marrubium vulgare L. (Lamiaceae) has a long history of use in traditional herbal medicine for the treatment of respiratory tract infections, inflammatory conditions, and pain. This study aimed to investigate the chemical composition, acute toxicity, and antinociceptive effects of the aqueous extract from *M. vulgare* leaves (AEMV). Antioxidant activity was evaluated using DPPH and reducing power assays. The chemical composition of AEMV was determined through LC–MS/MS, and the levels of total phenolics, flavonoids, and condensed tannins were quantified. Acute oral toxicity was assessed in male Swiss mice with a single oral dose of AEMV (1, 2, 5 g/kg). The analgesic impact

was examined through writhing, hot plate, and formalin tests. Our findings not only confirmed the safety of the extract in animal models but also revealed significant antioxidant activity in AEMV. High-performance liquid chromatography (HPLC) analysis identified important bioactive compounds, with marrubiin being a major component. Furthermore, AEMV demonstrated robust antinociceptive properties in all conducted tests, highlighting its potential as a valuable natural source of bioactive compounds suitable for a wide range of therapeutic applications.

Introduction

In recent decades, herbal medicines have garnered widespread acknowledgment worldwide for their effectiveness, affordability, and minimal side effects in disease management.^[1] Natural products, especially those from plants, have been instrumental as valuable reserves of crucial raw materials. These materials form the basis for discovering numerous bioactive molecules, providing a multitude of benefits in the fields of nutrition, cosmetics, and pharmaceuticals.^[2]

Among the medicinal plants documented among populations and enjoying a good therapeutic reputation that should undergo thorough investigations through chemical and biological deciphering, there is *Marrubium vulgare* L. This species is widely used in the Mediterranean area for its numerous therapeutic virtues.^[3] Traditional practitioners employ it against diabetes, respiratory tract infections, and disorders of bile secretion, acute benign bronchial infections, and colds.^[4]

Marrubium vulgare, commonly referred to as horehound and known as “Merriwa” in Morocco, emerges as the most prominent among the different *Marrubium* species. The noteworthy biological activities associated with *Marrubium*, as documented in the literature, are frequently attributed to its wide range of bioactive compounds, including flavonoids, diterpenoids, and phenylethanoid glycosides.^[5,6] Among these, Marrubiin, a well-known diterpenoid, was initially isolated from *M. vulgare* in 1842. This furanic labdane diterpene contributes, in part, to the therapeutic properties observed in various *Marrubium* species.^[7]

It has been documented that the *M. vulgare* plant showcases a variety of beneficial activities, including antioxidant, antifungal, hypoglycemic, and hypotensive properties.^[8] Moreover, the plant demonstrates antibacterial, antispasmodic, and insecticidal activities.^[9,10] Previously studies have highlighted additional attributes such as anticorrosion and vasorelaxant effects,^[11] positioning *M. vulgare* as a valuable source of bioactive compounds with analgesic properties.^[11] In Morocco, traditional healers currently use *M. vulgare* both independently and in combination with other herbs for its recognized vaso-relaxing, anti-hypertensive, analgesic, anti-inflammatory, anti-hepatotoxic, and anti-edematous activities.^[12,13,14] In this context, our study breaks new ground by investigating the therapeutic potential of *Marrubium vulgare* cultivated in Morocco, a departure from traditional reliance on wild-harvested sources. Additionally, our research delves into its chemical composition, providing a comprehensive analysis that adds a new dimension to our understanding of this medicinal plant. The present study is undertaken to experimentally evaluate the traditional uses of *M. vulgare* on animal models. In pursuit of this goal, we assessed the chemical composition, antioxidant, acute toxicity,

[a] Dr. A. Aitbaba, Dr. H. Kabdy, Dr. A. Baslam, Dr. H. Azraida, Prof. L. El Yazouli, Prof. Z. Sokar, Prof. A. Chait
Laboratory of Pharmacology, Neurobiology, Anthropobiology and Environment, Department of Biology, Faculty of Sciences Semlalia, Cadi Ayyad University, Marrakech, Morocco 511 – 40000

[b] Prof. R. Aboufatima
Laboratory of Biological Engineering, Faculty of Sciences and Technology, Sultan Moulay Slimane University, Beni Mellal, Morocco

[c] Prof. S. Garzoli
Department of Chemistry and Technologies of Drug, Sapienza University, P. le Aldo Moro, 5, 00185 Rome, Italy
E-mail: stefania.garzoli@uniroma1.it

© 2024 The Author(s). Chemistry & Biodiversity published by Wiley-VHCA AG. This is an open access article under the terms of the Creative Commons Attribution License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

Table 1. Mean values of total polyphenol, flavonoids, and tannin contents in the extract.			
	Total Polyphenol Content mg GAE/g DM	Flavonoids mg QE/g DM	Tanins mg CE/g DM
Crude extracts (mg/g)	232.6 ± 4.22	89.2 ± 2.07	26.94 ± 1.02

GAE: gallic acid equivalent, QE: Quercetin equivalent, CE: Catechin equivalent, DM: dry matter.

antinociceptive effects of the aqueous extract of *Marrubium vulgare* (AEMV).

Results and Discussion

Phytochemical Study

Our findings demonstrated that AEMV leaves contains significant and intriguing amounts of total polyphenols (232.6 ± 4.22 mg GAE/g), flavonoids (89.2 ± 2.07 mg QE/g), and condensed tannins (26.94 ± 1.02 mg CE/g) (Table 1). Comparing our results with others previously obtained for the same species collected but in different regions of the world, we observed many similarities and some differences that can be attributed to both intrinsic and extrinsic factors.^[15] Several studies conducted in Algeria have demonstrated slight variations in the content of polyphenols, flavonoids and condensed tannins.^[3,15] These differences could be explained by differences in the genotype, the environment, the plant organ and the type of solvent used to carry out the extraction process. In particular, the influences of climatic, edaphic and genetic factors on the biosynthesis and accumulation of total phenolic compounds and total flavonoids have been well defined.^[16,17]

AEMV was subjected to analysis by HPLC–PDA–MS/MS, where the phytoconstituents were identified based on their MS/MS fragments and retention times (Rt), revealing 10 major secondary metabolites comprising various phenolic compounds, flavonoids and diterpenes (Figure 1). The results were summarized in Table 2. Predominantly, marrubiin emerged as

the most abundant compound (Rt = 19.65 min). The others main constituents identified in the extract were kampherol glucoside (Rt = 21.94 min), quercetin diglucoside (Rt = 13.88 min) and the methyl gallate (Rt = 9.5 1 min). These compounds are known for their beneficial properties and may contribute to the pharmacological effects of the extract. Previous studies have suggested certain mechanisms of action of these compounds, primarily marrubiin, and their involvement in observed antioxidant and antinociceptive effects.^[18]

Antioxidant Activity

AEMV demonstrated antioxidant activity comparable to that of a standard antioxidant, as assessed through the DPPH and reducing power methods. The calculated IC₅₀ value was 1.53 ± 0.01 mg/mL, indicating its ability to convert the stable purple-colored DPPH radical into the yellow-colored DPPH-H form (Table 3). Furthermore, AEMV exhibited reducing power, with an IC₅₀ value of 1.83 ± 0.01 mg/mL. This is consistent with other previously reported studies.^[19,20] It is reported that the aqueous extract of *M. vulgare* leaves contained approximately 560 μmol/g of Trolox equivalent per gram of dry weight, which is an antioxidant used to mitigate damage from oxidative stress.^[21] Additionally, in a separate study, a methanol extract of the leaves exhibited potent antioxidant activity against DPPH, with an IC₅₀ value of 35 μg/mL for radical scavenging.^[22] The activity of extract was lower than the activity of the positive controls Quercetin and BHT. The difference in the DPPH scavenging activity and reducing power from one extract to another may

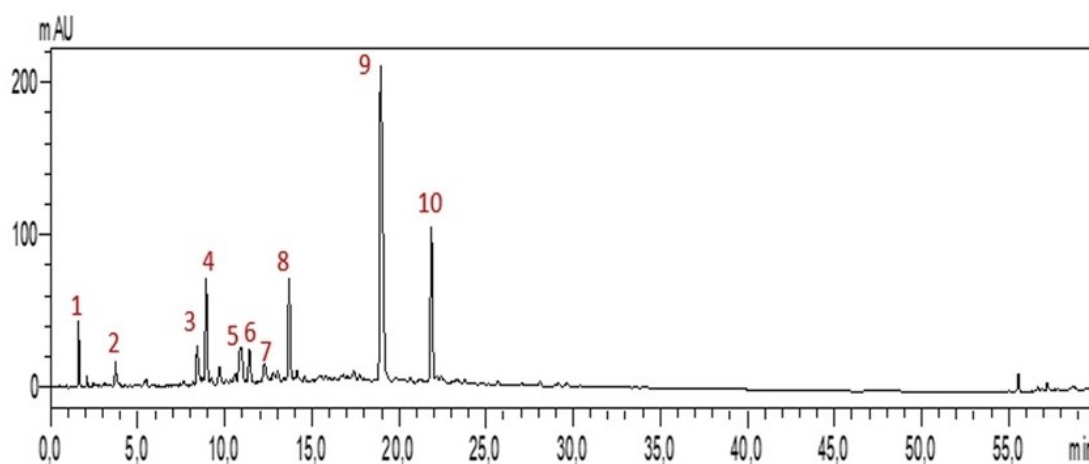


Figure 1. LC–MS/MS profile of *M. vulgare* extract.

Table 2. Annotated compounds from *Marrubium vulgare* aqueous extract using LC–MS/MS.

Peak No.	Rt(min)	[M–H] – (m/z)	MS/MS fragment	Proposed compound
1	1.83	331	125, 169	Gallic acid Glucoside
2	3.57	887	161, 179	Marruboside
3	8.28	153	109	3,4-Dihydrobenzoic acid
4	9.51	183	125	Methyl gallate
5	11.03	635	169, 331, 483	Tri-O-galloylglucose
6	11.67	191	111, 173	Quinic acid
7	12.40	179	108–135	Caffeic acid
8	13.88	625	301	Quercetin diglucoside
9	19.65	331	215, 221, 269	Marrubiin
10	21.94	447	285	Kampherol glucoside

Table 3. The antioxidant activity of the AEMV by DPPH and FRAP.

Antioxidant Assay	Plant Extract (IC ₅₀ = mg/mL)	Standard Antioxidant (IC ₅₀ = mg/mL)	
		Quercetin	BHT
DPPH	1.5 ± 0.0	0.1 ± 0.0	0.2 ± 0.0
FRAP	1.8 ± 0.0	0.1 ± 0.0	0.1 ± 0.0

Data are mean ± SEM.

be due to the difference in chemical composition and content of phenolic compounds.^[23]

Our *M. vulgare* extract showed a high content of total phenolic compounds thus suggesting the presence of potent antioxidants in the plant material. As demonstrated by Nagendrappa,^[24] phenolic compounds can directly contribute to antioxidant effects. Flavonoids, known for their robust antioxidant activities, have the ability to scavenge various reactive oxygen species (ROS) such as superoxide and nitric radicals.^[25]

Acute Toxicity

Estimation of LD₅₀

The oral administration of single doses (1000, 2000, and 5000 mg/kg body weight) of AEMV did not cause any mortality among the treated animals during the fourteen days of the test. There were no indications of toxicity. Additionally, no significant changes in body weight or relative organ weight were observed among the treated animals (Table 4, Figure 2). These findings suggest that the oral lethal dose 50 (LD₅₀) of AEMV was higher than 5000 mg/kg body weight in mice.

Hematological, Biochemical and Histopathological Analysis

The administration of AEMV did not alter the level of hematological (RBC, WBC and PLT) and biochemical parameters (Urea, uric acid, creatinine, ASAT, ALAT) at all doses (Figures 3, 4). Microscopic examination of the kidney, liver and spleen tissues showed a normal appearance similar to that of the control group, suggesting that no damaging changes or

Table 4. The relative organ weight of male mice treated with single oral doses of AEMV.

	Control	AEMV 1 g/kg	AEMV 2 g/kg	AEMV 5 g/kg
Body weight (g)	26.7 ± 1.19	26.02 ± 1.5	27.5 ± 0.9	26.9 ± 1.2
Relative organ weight (g/100 g bw)				
Liver	4.91 ± 0.51	6.10 ± 1.01	7.51 ± 0.66	6.14 ± 0.88
Right kidney	0.54 ± 0.04	0.54 ± 0.10	0.61 ± 0.10	0.60 ± 0.08
Left kidney	0.53 ± 0.03	0.52 ± 0.10	0.59 ± 0.10	0.61 ± 0.09
Spleen	0.51 ± 0.12	0.50 ± 0.08	0.70 ± 0.14	0.67 ± 0.15
Lungs	0.84 ± 0.06	0.76 ± 0.09	0.94 ± 0.03	1.03 ± 0.24
Brain	1.54 ± 0.03	1.5 ± 0.11	1.41 ± 0.05	1.67 ± 0.22
Heart	0.44 ± 0.04	0.42 ± 0.06	0.42 ± 0.02	0.42 ± 0.05
Stomach	0.95 ± 0.11	0.80 ± 0.15	0.97 ± 0.08	1.15 ± 0.20

Each value is represented as mean ± SD (n = 5). bw: body weight.

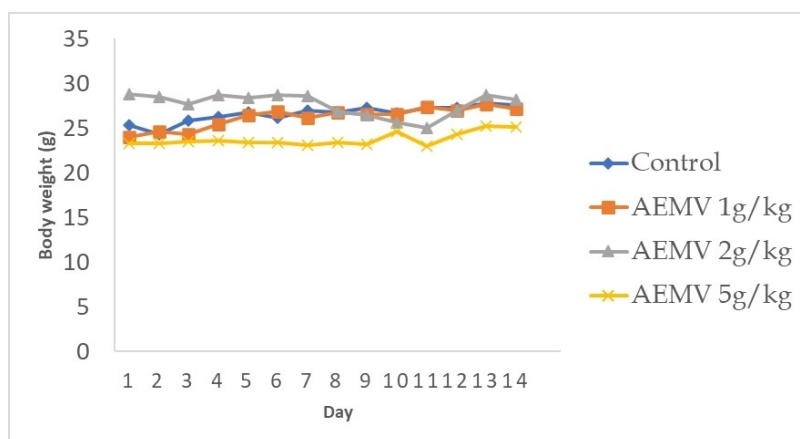


Figure 2. Evolution of the body weight of male mice ($n=5$) treated with a single dose of AEMV. No significant difference was recorded among groups ($p > 0.05$ versus control group). The data were analyzed by one-way ANOVA followed by Tukey's test.

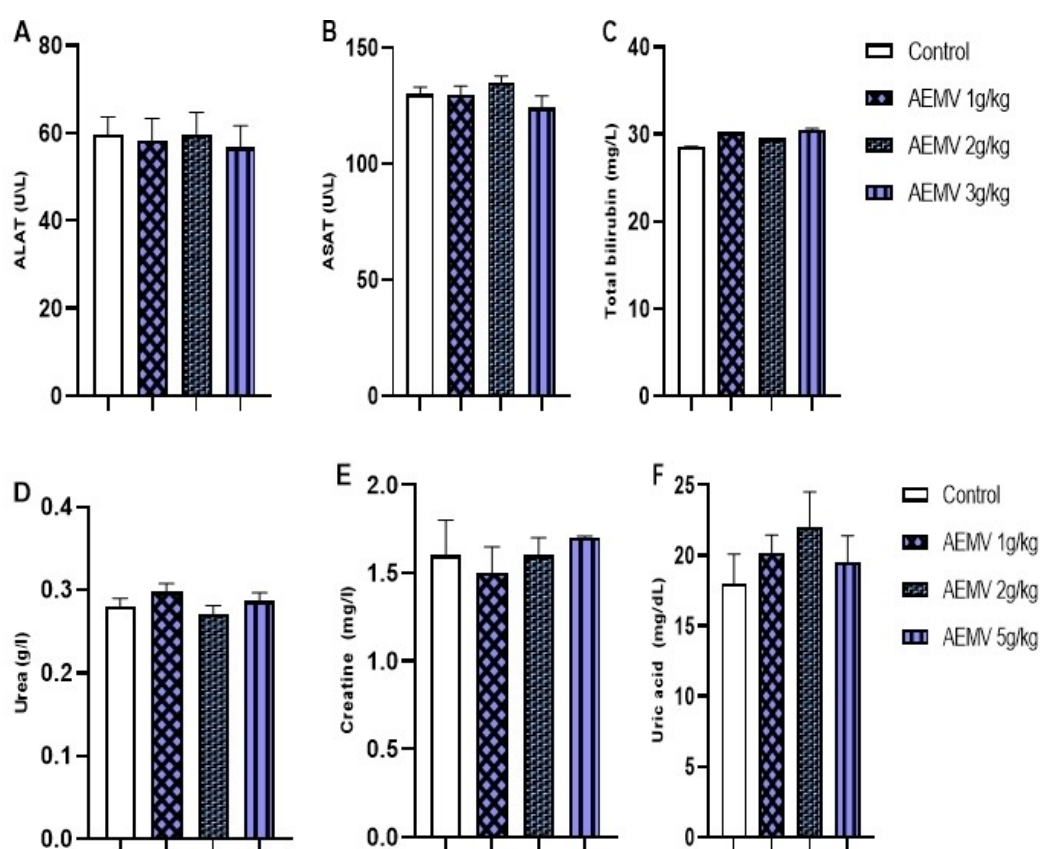


Figure 3. Effect of *Marrubium vulgare* aqueous extract (AEMV) on liver and renal functions: ALAT (A), ASAT (B), Total Bilirubine (C), Urea (D), Creatinine (E) and Uric Acid (F). Results are expressed as mean $\pm$ SEM ($n=5$ per group). Data were analyzed by one-way ANOVA followed by Tukey's post hoc test. There is no statically significance among all groups ($p > 0.05$).

morphological disturbances were caused by oral administration of the aqueous extract of *Marrubium vulgare* (Figure 5).

Marrubium vulgare is commonly utilized by the Moroccan population, even though with careful consideration, given its flavor and bitter taste, which raises questions regarding its potential toxicity. The initial phase of evaluating the effectiveness and safety of plant preparations often involves conducting acute toxicity tests in rodents. The insights gained from these

trials are invaluable for determining appropriate dosage levels for further pharmacological studies and for identifying any early indicators of toxicity. In this study, acute toxicity in mice was assessed following the OECD guideline. Our finding demonstrate the safety profile of the aqueous extract of Moroccan variety of *M. vulgare* leaves at a minimum dose of 5g/kg. According to Tohidi et al.^[18] substances with an oral LD₅₀ greater than 5 g/kg can be considered practically non-toxic. Therefore,

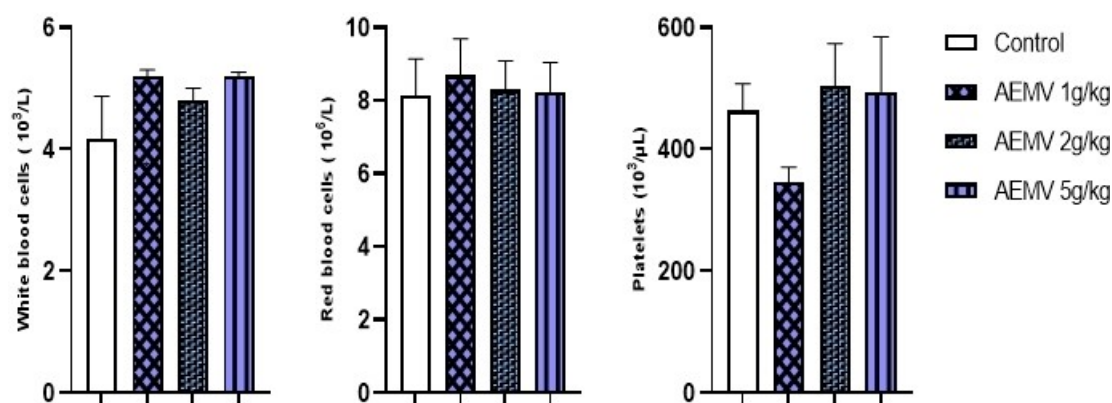


Figure 4. Effect of *Marrubium vulgare* aqueous extract (AEMV) on hematological samples: (A) White Blood cells, (B) Red Blood cells, and (C) Platelets. Results are expressed as mean $\pm$ SEM (n = 5 per group). Data were analyzed by one-way ANOVA followed by Tukey's post hoc test. There is no statically significance among all groups ($p > 0.05$).

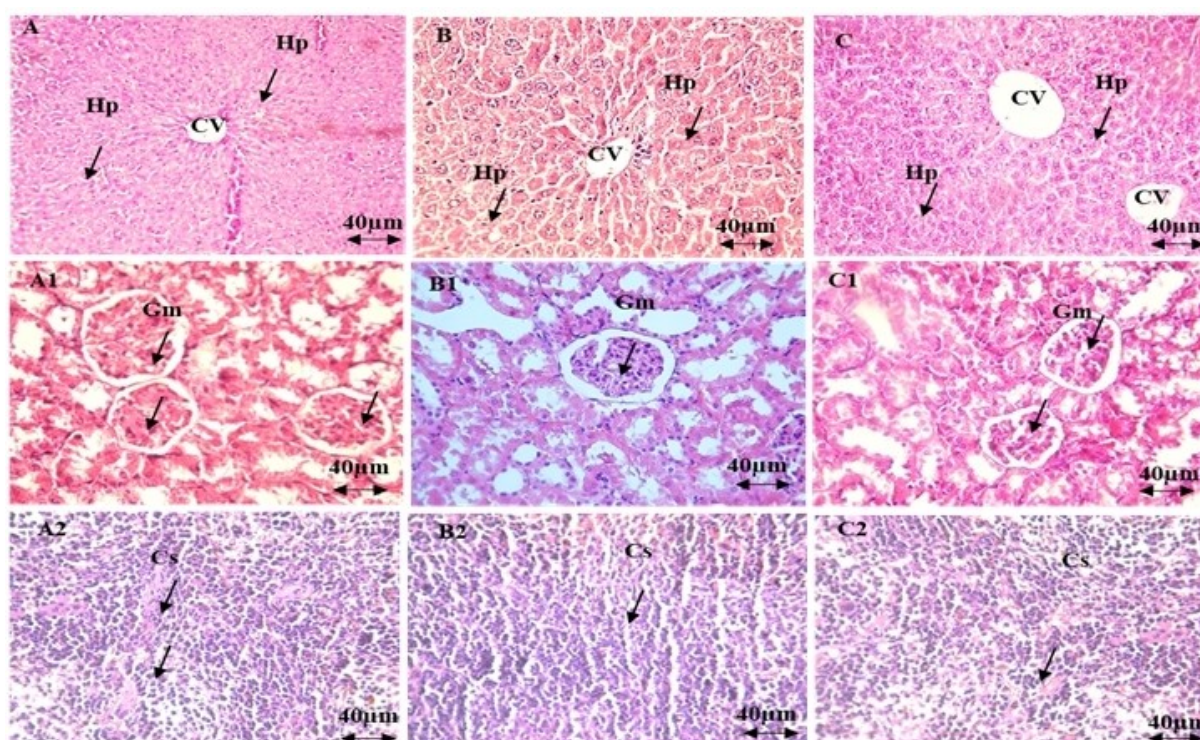


Figure 5. Histopathological examinations of organs (liver, kidneys, spleen) in acute toxicity. Kidney tissue: control (A, A1, A2), AEMV 2 g/kg (B, B1, B2) and AEMV 5g/kg (C, C1, C2). Sections were stained and colored with hematoxylin & Eosin ($\times 200$).

it can be suggested that the aqueous extract of *Marrubium vulgare* is devoid of acute oral toxicity.

Antinociceptive Activity

Hot Plate Test

The results of the analgesic effects of the plant extract and the standard drug are illustrated in Figure 6. Antinociceptive activity results reveal that AEMV demonstrated a significant increase ($p < 0.05$) in time latency at 30, 60, and 90 minutes when

compared to the control group. Similarly, the standard drug (morphine 10 mg/kg) significantly prolonged the time intervals in all cases compared to the control group.

Abdominal Writhing Test

The application of AEMV at all dosage levels resulted in a significant reduction in the number of abdominal constrictions compared to the control group (Figure 7). This observed effect was intensified at higher doses ($p < 0.01$) and was comparable

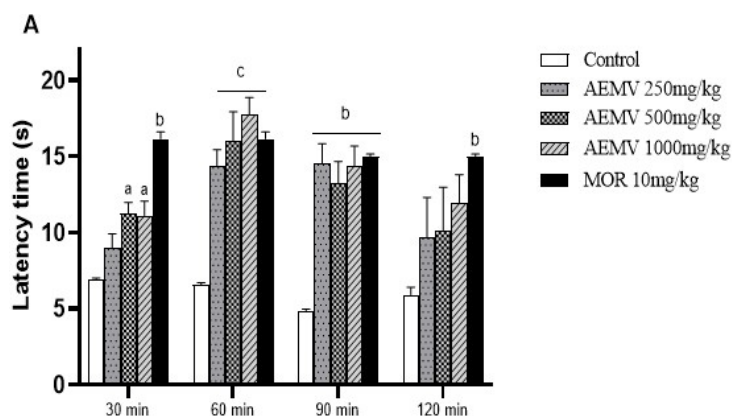


Figure 6. Effect of AEMV (250, 500, 1000 mg/kg) and MOR 10 mg/kg on nociception tests: Hot plate. The data were analyzed by one-way ANOVA followed by Tukey's test and results are presented as mean $\pm$ SEM. Superscript 'a' means p value < 0.05 ; 'b' means p value < 0.01 ; 'c' means p value < 0.001 compared to the control group.

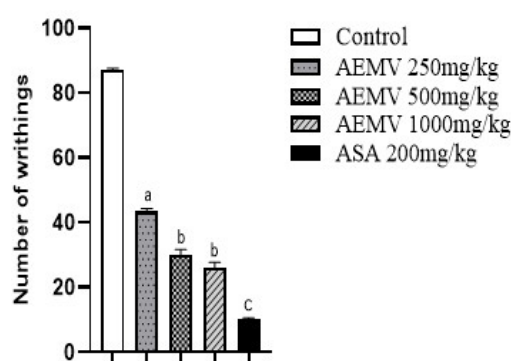


Figure 7. Effect of AEMV (250, 500, 1000 mg/kg) and ASA 200 mg/kg on the writhing test. The data were analyzed by one-way ANOVA followed by Tukey's test and results are presented as mean $\pm$ SEM. Superscript 'a' means p value < 0.05 ; 'b' means p value < 0.01 ; 'c' means p value < 0.001 compared to the control group.

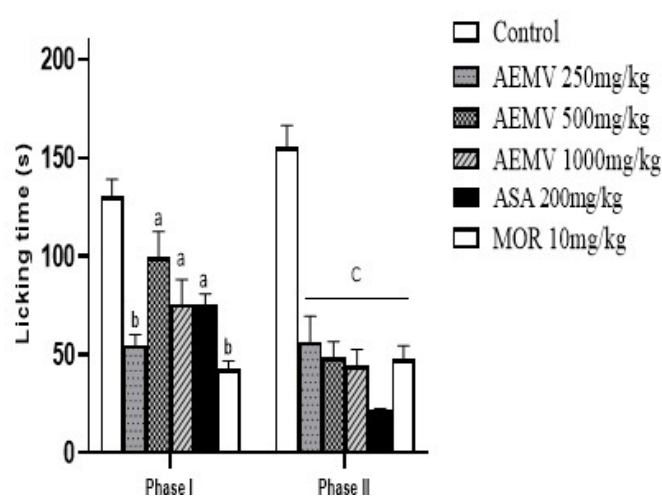


Figure 8. Effect of AEMV (250, 500, 1000 mg/kg) MOR 10 mg/kg and ASA 200 mg/kg on the formalin test. The data were analyzed by one-way ANOVA followed by Tukey's test and results are presented as mean $\pm$ SEM. Superscript 'a' means p value < 0.05 ; 'b' means p value < 0.01 ; 'c' means p value < 0.001 compared to the control group.

to the effects observed in the ASA group, which served as positive controls.

Formalin Test

The formalin test results are shown in Figure 8. The extract exhibited notable analgesic effects by decreasing paw licking time during both the neurogenic (Phase 1) and inflammatory (Phase 2) phases, with the reduction being particularly prominent in the latter phase. The administration of AEMV at doses of 250, 500, and 1000 mg/kg resulted in decreased in paw licking time during both phases of the formalin trial. In comparison, the reference drug ASA, demonstrated greater effectiveness in the second phase. Morphine, the standard drug, reduced licking time in both Phase 1 and Phase 2.

The aqueous extract of *M. vulgare* demonstrates potent pain-relieving effects across various tests, including the writhing, formalin, and hot plate tests. Specifically, the hot plate test results indicate significant analgesic activity of AEMV. Phytochemical screening of the extract reveals substantial amounts

of polyphenols, flavonoids, and tannins. Numerous studies employing various experimental animal models consistently show that flavonoids, tannins, and other polyphenolic compounds exhibit an analgesic effect.^[26,27] Abdominal contractions induced by intraperitoneal injection of acetic acid represent a method employed to investigate the peripheral analgesic effects of a substance. The pain resulting from acetic acid injection is attributed to the release of serotonin, histamine, bradykinin, substance P, and prostaglandins (PGE2 α , PGF2 α). These chemical mediators stimulate peripheral nociceptive neurons and increase vascular permeability.^[28,29] Abdominal contractions induced by acetic acid could be associated with sensitization of nociceptive receptors to prostaglandins. Therefore, it is plausible that the AEMV extract exerts its analgesic effect by likely inhibiting the synthesis or action of prostaglandins.^[30] Another potential activity of *M. vulgare* is its analgesic activity, which could prevent the release of inflamma-

tory mediators by peripheral action, as well as central antinociceptive effects.^[31] The analgesic activity of AEMV may be attributed to the presence of marrubiin and its derivatives. In line with this, research conducted by Stulzer et al.^[7] has demonstrated the beneficial impact of marrubiin against these chemical mediators, leading to pain inhibition.

The Formalin test serves as a suitable model for simulating pain in clinical conditions. In this test, subcutaneous injection of formalin induces a mild to moderate and sustained pain response attributed to tissue injury, manifesting in two distinct phases. The early acute phase, occurring 0–5 minutes after formalin injection, reflects centrally mediated pain associated with the direct stimulation of nociceptors and C fibers, along with the effects of substance P and bradykinin. Conversely, the chronic late phase, observed 15–60 minutes post-injection, mainly results from the secretion of inflammatory mediators such as serotonin, histamine, prostaglandins and bradykinin, coupled with the sensitization of central pain neurons. Centrally acting analgesics, such as opioids, exert control over both phases equally. In contrast, peripherally acting drugs, like corticosteroids and NSAIDs, specifically inhibit the late phase.^[29]

Our study showed that AEMV significantly reduced licking time in the neurogenic phase (Phase 1) of the formalin test compared to controls, indicating central pain suppression. Similarly, in the inflammatory phase (Phase 2), significant reductions in licking time were observed. These findings align with previous studies showing analgesic effects in both formalin-induced phases.^[11,22] This activity may be partly due to marrubiin's inhibitory action on formalin-induced pain.^[33] Additionally, Kanyonga et al.^[34] demonstrated that a methanol extract of *M. vulgare* exhibited analgesic activity comparable to acetylsalicylic acid. Previous studies also attribute the significant analgesic activity in the formalin test to marrubiin's inhibitory effects on formalin-induced pain.^[33] Immediate impacts on sensory and bradykinin receptors or via a glutamatergic pathway may contribute to the observed effect in the initial phase. Conversely, the antinociceptive effect in the later phase seems linked to inflammatory responses triggered by the arachidonic acid cascade.^[30,35] Additionally, results from the hot plate test support AEMV's central analgesic effect, significantly increasing response latency.

The hot plate test evaluates the reaction to a brief, intense stimulus resembling clinical pain. It assesses the intricate responses to a non-inflammatory, acute nociceptive signal and serves as a specific model for investigating solely central antinociceptive effects.^[36] Medications acting centrally trigger the release of endogenous peptides, which are conveyed to the spinal cord to hinder the transmission of pain impulses within the dorsal horn.^[37] Considering the AEMV capability to extend latency on the hot plate, we propose that this extract exhibits centrally mediated antinociceptive effects against thermally induced nociception. The study by Sousa et al.^[30] found no significant effect of the hydroalcoholic extract of *M. vulgare* on increasing latency in the hot plate test. However, this does not necessarily mean that the extract from *M. vulgare* has no central effect, as already determined by the formalin test. One possible explanation for this result could be the low efficacy of some

extracts or doses on thermal pain receptors, which may not translate into a significant increase in latency in the hot plate test.^[38] The hot plate test primarily assesses sensitivity to thermal pain and requires more robust analgesic effects to induce a noticeable increase in latency. In contrast, the formalin test evaluates pain induced by chemical irritation, for which the extract from *M. vulgare* showed central analgesic effects, explaining its ability to prolong latency in this test. Thus, while the extract may have central effects against pain, these effects may vary depending on the type of pain evaluated and the sensitivity of the receptors involved.

Conclusions

In conclusion, this study sheds light on the potential therapeutic benefits of *Marrubium vulgare* L., a plant with a rich history in traditional medicine. Through comprehensive analysis, the study explored the phytochemical composition of *M. vulgare* L. leaves using HPLC-MS/MS, revealing significant bioactive compounds, including marrubiin. Antioxidant activity was assessed, highlighting substantial potential. Our findings not only affirm the safety of the extract in animal models but also underscore its notable antinociceptive properties, emphasizing the extract's pharmacological significance. Moving forward, further exploration of *M. vulgare* may unlock novel therapeutic avenues, offering valuable contributions to healthcare and pharmaceutical industries. Additionally, further research and investigations are essential to delve into the specific mechanisms of action and assess the extract's efficacy in both preclinical and clinical contexts.

Experimental Section

Chemicals Drugs

Morphine (MOR), indomethacin and acetylsalicylic acid (ASA) were obtained from Laprophan in Morocco. Formalin, acetic acid, carrageenan and xylene were sourced from Sigma-Aldrich in France. Before application, all solutions, drugs and chemicals were freshly prepared and administered at a dosage of 10 mL/kg.

Plant Material

The leaves of *M. vulgare* were collected in the Ourika region, located 30 km southeast of Marrakech, Morocco (31°17'45" N, 7°42'36" W). The initial verification of the plant authenticity was conducted by botanist Professor A. Ouhammou and pharmacologist Professor A. Chait from the Faculty of Sciences at Semlalia, Cadi Ayyad University. Subsequently, the plants were meticulously preserved and documented under voucher specimen MARK-14374 in the herbarium of the Department of Biology, Faculty of Sciences Semlalia, Cadi Ayyad University, Marrakech, Morocco.

Extract Preparation

The powder was prepared by finely grinding thoroughly washed and dried leaves. Next, 50 g of the powdered material underwent extraction with distilled water at a ratio of 1g to 10 mL, with

continuous agitation for 12 hours. The resulting aqueous macerate underwent centrifugation at 1200 rpm, followed by filtration and lyophilization using a Christ instrument. The lyophilized powder was then stored under aseptic conditions in amber bottles at 4 °C until required for use. The process yielded a 17.30% extraction yield.

Phytochemical Study

Total Phenolic Content (TPC)

The TPC of the extract was determined using a modified Folin-Ciocalteu method.^[39] To start the assessment, a 0.4 mL aliquot of the diluted extract was blended with 1.5 mL of the Folin–Ciocalteu reagent. Subsequently, 1.6 mL of a 7.5% sodium carbonate solution was added. The resulting mixture was then kept in darkness at room temperature for 2 hours, and the absorbance was measured at 765 nm. Gallic acid served as the standard reference, and the findings were expressed as milligrams of gallic acid equivalent per gram of dry weight (mg GAE/g DW).

Total Flavonoid Content (TFC)

The quantification of TFC in the extract was conducted using a method outlined in a previous study.^[40] In this procedure, a 200 μ L extract was first diluted with 1 mL of distilled water. Following this, 60 μ L of 5% NaNO₂ and 60 μ L of 10% AlCl₃ were added to the solution. After a five-minute incubation period, 400 μ L of 1 M NaOH was introduced into the mixture. Subsequently, the absorbance was measured at 510 nm, and the TFC was expressed as milligrams of catechin equivalents per gram of dry matter (mg CE/g DM).

Total Condensed Tannins

The determination of condensed tannins content was carried out using a previously described method.^[41,42] In a concise overview, the diluted samples (400 μ L) were mixed with 3 mL of a 4% methanol vanillin solution and 1.5 mL of concentrated HCL. After 15-minute incubation, the absorbance was measured at 500 nm. The total condensed tannin content was subsequently quantified and expressed as milligrams of catechin equivalent per gram of dry matter (mg CE/g DM).

LC–MS/MS Analysis

Phytochemical analysis of AEMV was carried out via HPLC with Photodiode Array Detection-Tandem Mass Spectrometer. A Shimadzu LC/MS 8050 system (Shimadzu, Japan) equipped with a triple quad spectrometer with an ESI source, was used. Separation was accomplished using a C18 reversed-phase column (Zorbax Eclipse XDB-C18, rapid resolution, 4.6×150 mm, 3.5 μ m, Agilent, Santa Clara, CA, USA). The elution involved a gradient of water and acetonitrile (ACN) (both containing 0.1% formic acid), ranging from 5% to 30% ACN over 60 minutes, with a flow rate of 1 mL/min. Sample injection was automated using an autosampler SIL-40C xs. The LC solution software (Shimadzu, Japan) controlled the instrument, and the MS operated in the negative ion mode.

Antioxidant Activity

Radical Scavenging Activity (DPPH)

The antioxidant activity of the extract was assessed using the 2,2-diphenyl-1-picrylhydrazyl radical (DPPH) assay, as previously described by Mansouri et al.,^[43] with specific modifications.^[44] Summarily, 1.5 mL of a methanolic DPPH solution (6×10^{-5} M) was combined with 60 μ L of the extract at varying concentrations (1, 2, 4, 6, and 8 mg/mL) of AEMV. The mixture was then protected from light and allowed to stand at room temperature for 3 minutes. After this incubation period, absorbance was measured at 515 nm, using a negative control consisting of 1.5 mL of the DPPH solution and 60 μ L of methanol. The study also incorporated positive controls, such as butylated hydroxytoluene (BHT) and quercetin. The percentage of inhibition was calculated using the formula: % Inhibition = [(A control – A sample)/A control] × 100

Where A control represents the absorbance of the control, and A sample denotes the absorbance of the test compound.

The concentration of the sample that causes 50% inhibition (IC₅₀) was ascertained from a graph illustrating the percentages of inhibition plotted against the sample concentrations.

Ferric Reducing Ability Power (FRAP)

The FRAP test, conducted following the Oyaizu method,^[45] involves inhibiting the formation of Fe (II)-Ferrazine complexes during sample incubation with ferrous iron. Practically, a mixture of distilled water (1 mL), phosphate buffer (2.5 mL; 0.2 M; pH = 6.6), and potassium ferricyanide (K₃[Fe (CN)₆]) (2.5 mL; 1%) was combined with 0.5 mL of solutions of various strengths. After a 30-minute incubation, 2.5 mL of distilled water, 2.5 mL of 10% trichloroacetic acid, and 0.5 mL of FeCl₃ were added to the mixture. Absorption was measured at 700 nm. Quercetin and butylated hydroxytoluene (BHT) served as positive controls.

Animal Study

Animals

Adult male Swiss mice (25–35 g) and Wistar albino rats (150–200 g) were obtained from the animal care unit of the Faculty of Science Semlalia, Cadi Ayyad University, Marrakech, Morocco. These animals were housed under consistent ambient temperature conditions (22 ± 2 °C) with a 12-hour light/12-hour dark cycle and had unrestricted access to food and water. All animal procedures strictly complied to the guidelines outlined in the European Council Directive (EU2010/63). The study was conducted ethically and received approval from the Council Committee of the Research Laboratory, Faculty of Science, Cadi Ayyad University of Marrakech. Every possible measure was taken to minimize potential animal suffering.

Acute Toxicity

The acute toxicity study's limit test dose (5000 mg/kg) was conducted following the guidelines of the Organization for Economic Cooperation and Development (OECD) (guideline n. 423). Four groups of mice (24 mice in total), each consisting of six animals, were employed. Three groups received oral administration of AEMV at doses of 1000, 2000, and 5000 mg/kg, while the remaining group (negative control) received distilled water. Administration was carried out at a rate of 10 mL/kg. The mice were

closely observed for signs of toxicity and mortality during the initial two hours following the administration of the extract. Body weights of the mice were recorded daily for a period of 14 days.

Biochemical, Hematological and Histological Analyses

On day 14, the animals underwent euthanasia via cervical dislocation to obtain blood samples for the assessment of potential changes in hematological and biochemical parameters. Organs, including the liver, spleen, and kidneys, were preserved in buffered formalin (10%) and subsequently embedded in paraffin. Following this, 5 μ m sections were obtained using a microtome and subjected to hematoxylin and eosin staining for microscopic examination. The relative organ weight of each animal was calculated as follows:

Relative organ weight

$$= \left[\frac{\text{Absolute organ weight (g)}}{\text{Body weight of the animal on sacrifice day (g)}} \right] \times 100$$

Analgesic Tests

Hot Plate Test

The methodology previously employed by Okolo et al.^[46] was implemented in this study. The animals were subjected to a test environment comprising a glass cylinder placed on a heated metal plate maintained at a temperature of $55 \pm 1^\circ\text{C}$. The reaction time was gauged by measuring the latency to unpleasant responses such as licking, shaking one paw, and jumping. The treated mice received three oral doses of AEMV (250, 500, and 1000 mg/kg), while the control group was orally administered 10 mL/kg of water. Additionally, an intraperitoneal injection of 10 mg/kg of morphine was administered. After treatment, latencies to nociceptive responses were measured at intervals of 30, 60, 90, and 120 minutes.

Abdominal Writhing Induced by Acetic Acid

The procedure outlined by Koster et al.^[47] was employed in this study. In summary, intra-peritoneal injections of 0.6% acetic acid (0.1 mL/10 g) were administered to mice. Thirty minutes before the injections, all experimental groups (5 mice per group) were treated with either the vehicle (v.o.), ASA (200 mg/kg, i.p.), or the three doses of the AEMV (250, 500, 1000 mg/kg, v.o.). Subsequently, individual cages were designated for each group to facilitate counting of abdominal spasms over a 30-minute period.

Paw Licking Induced by Formalin

The experiment adhered to the methodology described by Hunskaar & Hole.^[48] Mouse groups ($n=5$) were orally administered various doses of AEMV (250, 500, and 1000 mg/kg, v.o.) 45 minutes before receiving a 20 μ L injection of 2% formalin (v/v in 0.9% saline) into the subplantar area of the right hind paw. The control group received a vehicle (10 mL/kg of saline, v.o.). The reference analgesic medications, morphine (10 mg/kg, i.p.) and ASA (200 mg/kg, i.p.), were included. Subsequent to formalin injection in each mouse, the duration of paw licking was measured during two intervals: 0–5 min (neurogenic phase) and 15–30 min (inflammatory phase).

Statistical Analysis

Each measurement's data were presented as mean values $\pm$ SEM (standard error of the mean). The analysis and presentation of results were conducted using SigmaPlot 12.0 software, employing one-way analysis of variance (one-way ANOVA). In cases where differences were deemed significant ($p < 0.05$), post hoc analysis techniques were applied.

Funding

This study did not receive financial support.

Ethical Issue

Reducing the number of animals and suffering were done according to the recommendations of the European Council Directive (EU2010/63), and authorized by the council committee of the faculty of Sciences, Cadi Ayyad University Marrakech Morocco.

Author Contributions

A. Aitbaba: Conceptualization, Data curation, Formal analysis, Writing-original draft preparation, Validation; H. Kabdy: Data curation, Validation, Writing-original draft preparation; A. Baslam: Data curation, Writing-original draft preparation; H. Azraida: Data curation; R. Aboufatima: Formal analysis, Validation; L. Elyazouli: Formal analysis, Validation; Z. Sokar: Conceptualization; S. Garzoli: Writing-review and editing, Supervision; A. Chait: Conceptualization, Writing-review and editing, Supervision.

Acknowledgements

The authors express our gratitude to M. A. Regragui for his assistance in handling the animals. Open Access publishing facilitated by Università degli Studi di Roma La Sapienza, as part of the Wiley - CRUI-CARE agreement.

Conflict of Interests

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords: *Marrubium vulgare* L. • Chemical Characterization • LC-MS/MS • Toxicity • Biological Effects

- [1] M. Ekor, *Front. Pharmacol.* **2014**, *4*, <https://www.frontiersin.org/articles/10.3389/fphar.2013.00177>.
- [2] J.-Y. Chabrier, **2010**, <https://hal.univ-lorraine.fr/hal-01739123>.
- [3] N. Amessis-Ouchemoukh, I. M. Abu-Reidah, R. Quirantes-Piné, K. Madani, A. Segura-Carretero, *Ind. Crops Prod.* **2014**, *61*, 120–129. <https://doi.org/10.1016/j.indcrop.2014.06.049>.
- [4] M. Ćimović, K. Jeremić, N. Salaj, N. Gavarić, B. Kiprovski, V. Sikora, T. Zeremski, *Molecules* **2020**, *25*(12), 2898. <https://doi.org/10.3390/molecules25122898>.
- [5] F. Piozzi, M. Bruno, S. Rosselli, A. Maggio, *Nat. Prod. Commun.* **2006**, *1*(7), 585–592. <https://doi.org/10.1177/1934578x0600100713>.
- [6] D. Rigano, C. Formisano, A. Basile, A. Lavitola, F. Senatore, S. Rosselli, M. Bruno, *Phytother. Res.* **2007**, *21*(4), 395–397. <https://doi.org/10.1002/ptr.2061>.
- [7] H. K. Stulzer, M. P. Tagliari, J. A. Zampirolo, V. Cechinel-Filho, V. Schlemper, *J. Ethnopharmacol.* **2006**, *108*(3), 379–384. <https://doi.org/10.1016/j.jep.2006.05.023>.
- [8] M. Rezgui, N. Majdoub, B. Mabrouk, A. Baldisserotto, A. Bino, L. B. Ben Kaab, S. Manfredini, *Journal de Mycologie Médicale* **2020**, *30*(1), 100927. <https://doi.org/10.1016/j.mycmed.2020.100927>.
- [9] A. Béjaoui, I. B. Salem, A. Boulila, *J. Essent. Oil Bear Pl.* **2017**, *20*(5), 1244–1253. <https://doi.org/10.1080/0972060X.2017.1377118>.
- [10] V. Mittal, A. Nanda, *Pharm. Biol.* **2017**, *55*(1), 1337–1347. <https://doi.org/10.1080/13880209.2017.1297837>.
- [11] C. Meyre-Silva, R. A. Yunes, V. Schlemper, F. Campos-Buzzi, V. Cechinel-Filho, *Farm. J. Home* **2005**, *60*(4), 321–326. <https://doi.org/10.1016/j.farmac.2005.01.003>.
- [12] B. Ahmed, M. H. Masoodi, A. H. Siddique, S. Khan, *Nat. Prod. Res.* **2010**, *24*(18), 1671–1680. <https://doi.org/10.1080/14786410802280976>.
- [13] A. Boudjelal, C. Henchiri, L. Siracusa, M. Sari, G. Ruberto, *Fitoterapia* **2012**, *83*(2), 286–292. <https://doi.org/10.1016/j.fitote.2011.11.005>.
- [14] S. El Bardai, N. Morel, M. Wibo, N. Fabre, G. Llabres, B. Lyoussi, J. Quetin-Leclercq, *Planta Med.* **2023**, *69*(1), 75–77. <https://doi.org/10.1055/s-2003-37042>.
- [15] I. Eseberri, J. A. Trepiana, A. L. Rodríguez, A. I. Gómez García, H. Carr Ugarte, M. González, M. P. Portillo Baquedano, *Nutrients* **2022**, *14*(9), <https://doi.org/10.3390/nu14091925>.
- [16] K. Bouterfas, Z. Mehdadi, D. Benmansour, M. B. Khaled, M. Bouterfas, A. Latreche, *Eur. J. Org. Chem.* **2014**, *4*(5), 292–308. <https://doi.org/10.4236/ijoc.2014.45032>.
- [17] R. Kiani, A. Arzani, S. A. M. Mirmohammady Maibody, *Front. Plant Sci.* **2021**, *12*, 646221. <https://doi.org/10.3389/fpls.2021.646221>.
- [18] B. Tohidi, M. Rahimmalek, A. Arzani, *Food Chem.* **2017**, *220*, 153–161. <https://doi.org/10.1016/j.foodchem.2016.09.203>.
- [19] G. L. Kennedy Jr., R. L. Ferenz, B. A. Burgess, *J. Appl. Toxicol.* **1986**, *6*(3), 145–148. <https://doi.org/10.1002/jat.2550060302>.
- [20] A. Boulila, A. Sanaa, I. B. Salem, N. Rokbeni, Y. M'rabet, K. Hosni, X. Fernandez, *Ind. Crops Prod.* **2015**, *76*, 616–622. <https://doi.org/10.1016/j.indcrop.2015.07.069>.
- [21] F. Yahiaoui, M. Zaouani, M. Kardjadj, A. Laghouati, R. Kadour, N. Bouzid, M. H. Ben-Mahdi, *Phytothérapie* **2018**, *16*(S1), S32–S39. <https://doi.org/10.3166/phyto-2018-0015>.
- [22] S. A. Dar, A. Bhushan, P. Gupta, *Botanical Leads for Drug Discovery* **2020**, 255–275. https://doi.org/10.1007/978-981-15-5917-4_12.
- [23] A. Chedia, H. Ghazghazi, H. Brahim, M. Abderrazak, *Tunis. J. Med. Plants Nat. Prod.* **2014**, *11*, 73–79.
- [24] R. Sudan, M. Bhagat, S. Gupta, J. Singh, A. Koul, *BioMed Res. Int.* **2024**, *2014*, <https://www.hindawi.com/journals/bmri/2014/179865/abs/>.
- [25] G. Nagendrappa, *An appreciation of free radical chemistry 3. Free radicals in diseases and health*, Resonance, vol. 10, p. 65–74, **2005**.
- [26] B. Halliwell, *Free Radic. Biol. Med.* **2009**, *46*(5), 531–542.
- [27] R. Gomathi, S. Manian, *Ind. Crops Prod.* **2015**, *73*, 134–143. <https://doi.org/10.1016/j.indcrop.2015.03.090>.
- [28] S. Saravanan, K. Arunachalam, T. Parimelazhagan, *Ind. Crops Prod.* **2014**, *54*, 272–280. <https://doi.org/10.1016/j.indcrop.2014.01.038>.
- [29] Y.-F. Chen, H.-Y. Tsai, T.-S. Wu, *Planta Med.* **1995**, *61*(1), 2–8. <https://doi.org/10.1055/s-2006-957987>.
- [30] A. Tjølsen, O.-G. Berge, S. Hunskaar, J. H. Rosland, K. Hole, *Pain* **1992**, *51*(1), 5–17. [https://doi.org/10.1016/0304-3959\(92\)90003-T](https://doi.org/10.1016/0304-3959(92)90003-T).
- [31] M. M. de Souza, R. A. P. de Jesus, V. Cechinel-Filho, V. Schlemper, *Phytomedicine* **1998**, *5*(2), 103–107. [https://doi.org/10.1016/S0944-7113\(98\)80005-6](https://doi.org/10.1016/S0944-7113(98)80005-6).
- [32] A. Bouyahya, F. E. Guaouguaou, N. El Omari, N. El Menyiy, A. Balahbib, M. El-Shazly, Y. Bakri, *J. Pharm. Anal.* **2022**, *12*(1), 35–57. <https://doi.org/10.1016/j.jpha.2021.07.004>.
- [33] N. Zaabat, A.-E. Hay, S. Michalet, N. Darbour, C. Bayet, I. Skandrani, L. Chekir-Ghedira, S. Akkal, M.-G. Dijoux-Franca, *Food Chem. Toxicol.* **2011**, *49*(12), 3328–3335. <https://doi.org/10.1016/j.fct.2011.08.026>.
- [34] R. A. P. De Jesus, V. Cechinel-Filho, A. E. Oliveira, V. Schlemper, *Phytomedicine* **2000**, *7*(2), 111–115. [https://doi.org/10.1016/S0944-7113\(00\)-80082-3](https://doi.org/10.1016/S0944-7113(00)-80082-3).
- [35] P. M. Kanyonga, M. A. Faouzi, B. Meddah, M. Mpona, E. M. Essassi, Y. Cherrah, *J. Chem. Pharm. Res.* **2011**, *3*(1), 199–204.
- [36] D. Dubuisson, S. G. Dennis, *Pain* **1977**, *4*, 161–174. [https://doi.org/10.1016/0304-3959\(77\)90130-0](https://doi.org/10.1016/0304-3959(77)90130-0).
- [37] R. K. Sutradhar, A. M. Rahman, M. Ahmad, B. S. Chandra, A. Saha, S. K. Guha, *Iran. J. Pharmacol.* **2006**, *5*, 175–178. https://www.sid.ir/EN/VEWSSID/J_pdf/101020060205.pdf.
- [38] G. F. Ibrionke, K. I. Ajiboye, *Int. J. Pharmacol.* **2007**, *3*(1), 111–115.
- [39] V. L. Singleton, R. Orthofer, R. M. Lamuela-Raventós, *Analysis of total phenols and other oxidation substrates and antioxidants by means of folin-ciocalteu reagent. In Methods in Enzymology* **1999**, *299*, 152–178. Elsevier. [https://doi.org/10.1016/S0076-6879\(99\)99017-1](https://doi.org/10.1016/S0076-6879(99)99017-1).
- [40] J. Zhishen, T. Mengcheng, W. Jianming, *Food Chem.* **1999**, *64*(4), 555–559. [https://doi.org/10.1016/S0308-8146\(98\)00102-2](https://doi.org/10.1016/S0308-8146(98)00102-2).
- [41] A. Aitbaba, Z. Sokar, A. Chait, *Bangladesh J. Pharmacol.* **2023**, *18*(2), 41–48.
- [42] D. Heimler, P. Vignolini, M. G. Dini, A. Romani, *J. Agric. Food Chem.* **2005**, *53*(8), 3053–3056. <https://doi.org/10.1021/jf049001r>.
- [43] A. Mansouri, G. Embarek, E. Kokkalou, P. Kefalas, *Food Chem.* **2005**, *89*(3), 411–420. <https://doi.org/10.1016/j.foodchem.2004.02.051>.
- [44] A. Baslam, A. Aitbaba, R. Aboufatima, F. Agouram, S. Boussaa, A. Chait, M. Baslam, *Horticulturae* **2023**, *9*(11), 11. <https://doi.org/10.3390/horticulturae9111196>.
- [45] M. Oyaizu, *Japanese J. Nutr. Diet.* **1986**, *44*(6), 307–315.
- [46] C. O. Okolo, P. B. Johnson, E. M. Abdurahman, I. Abdu-Aguye, I. M. Hussaini, *J. Ethnopharmacol.* **1995**, *45*(2), 125–129. [https://doi.org/10.1016/0378-8741\(94\)01199-A](https://doi.org/10.1016/0378-8741(94)01199-A).
- [47] R. Koster, M. Anderson, E. De Beer, *Medicine* **1959**.
- [48] S. Hunskaar, K. Hole, *Pain* **1987**, *30*(1), 103–114. [https://doi.org/10.1016/0304-3959\(87\)90088-1](https://doi.org/10.1016/0304-3959(87)90088-1).

Manuscript received: January 25, 2024

Version of record online: April 17, 2024

[Correction added on 24 February 2025, after first online publication: The copyright line was changed.]